

TRM V3.1–V3.2 Addendum: From Memory Action-Closure to Lattice-Derived Minimal Action

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Technical Addendum to the TRM/TQM V3.0 Review Baseline
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This document is a compact, citable supplement to the V3.0 trilogy, not a new major paper generation.

1 Abstract

This addendum provides a claim-safe update to the TRM V3.0 paper trilogy. V3.1 strengthens Gap 1 to an *action-derived memory-closure candidate*, and V3.2 strengthens the action layer to a *minimal effective action that is plausibly lattice-derived under comparative guard tests*. These results improve structural and comparative support while remaining below theorem-level first-principles closure.

2 Relation to V3.0 Review Baseline

V3.0 established the public review baseline with broad weak-field tested-effective guard coverage. It did not claim theorem-level microscopic closure and did not claim GR replacement. V3.1 and V3.2 are targeted hardening layers on top of that baseline, not a replacement of the baseline claim boundary and not a “V4” reset.

3 V3.1: Action-Derived Memory-Closure Candidate

V3.1 focuses on Gap 1 and hardens the memory-channel path:

$$A_{\text{dyn}} \propto \phi \rightarrow A_{\text{dyn}}^2 |\dot{\mu}| \rightarrow \phi^2 |\dot{\mu}|.$$

Status in V3.1:

- Gap 1 is strengthened to an *action-derived memory-closure candidate*.
- The path is guard-supported and variation-compatible.
- It remains *not theorem-level first-principles closure*.

4 V3.2: Minimal Effective Action from TQM Lattice/Synchronization Principles

V3.2 moves one level deeper by testing whether the minimal effective action itself is plausibly motivated by TQM lattice/synchronization structure.

Status in V3.2:

- The minimal effective action is *plausibly lattice-derived under comparative guard tests*.
- This remains a candidate-level derivation status, not theorem-level closure.

5 Guard Summary

V3.2 guard block (UA16–UA23), in compact form:

- **UA16:** lattice energy reduces to minimal scalar action in tested weak-field cases.
- **UA17:** $A_{\text{dyn}}^2 \kappa$ is preserved under coarse-graining.
- **UA18:** V3.1 UF13–UF15 behavior is reproduced without retuning.
- **UA19:** non-minimal terms remain penalized/subleading.
- **UA20:** MC/FD/TO limits remain preserved.
- **UA21:** alternative lattice-energy forms degrade minimal-action closure quality.
- **UA22:** minimal-action closure remains robust under controlled lattice perturbations.
- **UA23:** non-minimal lattice actions do not outperform the minimal action under penalty-aware comparison.

Combined interpretation: the framework supports not only positive-fit behavior of one action form, but comparative preference and robustness for the minimal action class in the tested regime.

6 Remaining Theorem-Level Gaps

The following remain open:

1. theorem-level first-principles microscopic derivation of the minimal effective action,
2. theorem-level uniqueness/necessity of the selected leading interaction hierarchy,
3. full theorem-level closure across remaining first-principles gaps.

7 Conclusion

Relative to V3.0, V3.1 and V3.2 provide materially stronger claim-safe support for the memory/action layer:

- V3.1: action-derived memory-closure candidate,
- V3.2: minimal effective action plausibly lattice-derived under comparative guards.

Boundary remains explicit:

- no GR replacement claim,
- no theorem-level first-principles closure claim.